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Palaeomagnetic evidence for southward displacement of the Chuan Dian fragment of the Yangtze Block

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SUMMARY

Cretaceous red sandstones of the Feitianshan Formation and the Xiaoba Formation were sampled at 33 sites from the Dadeli and Mishi synclines of Xichang (27.9°N, 102.3°E). The study area is a part of the Chuan Dian fragment bounded by the Xianshuihe–Xiaojiang and the Red River fault systems, which in turn constitute the southwestern part of the Yangtze Block. Almost all the samples give a characteristic palaeomagnetic direction with unblocking temperatures up to 680°C. The primary nature of magnetization is ascertained by a positive fold test with a 99 per cent confidence level for the Dadeli Syncline. The tilt corrected overall mean direction of the 33 sites is $D = 3.7^\circ$, $I = 41.5^\circ (\alpha_{95} = 3.4^\circ)$, with a corresponding palaeopole at 85.2°N, 241.7°E($A_{95} = 3.5^\circ$). This pole occupies the near-sided position with respect to the estimate of the Cretaceous pole of the Sichuan Basin, indicating that the Xichang area experienced a significant southward displacement. Combined with earlier reported palaeomagnetic data from the Chuan Dian fragment, a significant southward displacement of $6.7^\circ \pm 3.5^\circ$ in latitude is estimated for the whole fragment with respect to the Sichuan Basin since the Late Cretaceous. Extrusion dynamics in the Asian continent due to its collision with India brought about the southward displacement of the Chuan Dian fragment. Declination data indicate that the southern part of the Chuan Dian fragment was subjected to clockwise sense rotation of up to 45°. This significantly large tectonic rotation probably occurred during extrusion of this fragment from the north.

Key words: Chuan Dian fragment, deformation, displacement, palaeomagnetism, tectonics, Yangtze Block.

1 INTRODUCTION

Mesozoic to Cenozoic tectonics in southern Asia is dominated by the collision of several tectonic blocks with the Asian continent. The suturing of the Lhasa Block with the Qiangtang Block in the Jurassic was followed by India–Asia collision in the early Tertiary (Allegre *et al.* 1984; Rowley 1996). The Indian subcontinent subsequently penetrated into southern Asia and continued to move northwards. Collision and penetration of tectonic blocks resulted in large-scale deformation in the Asian continent.

The mode of tectonic deformation of the Asian continent has been widely debated over the last two decades. The Asian continent is regarded as behaving either as a deforming continuum material or as an assemblage of rigid bodies. In the continuum model, material penetrated into Asia due to the intrusion of India is absorbed by crustal thickening beneath the Tibetan Plateau and by tectonic rotation of crust along both sides of India (Houseman & England 1986, 1993). The rigid body model stresses that major strike-slip faults play an important role in accommodating the deformation between India and Asia, and that the regions between the major faults move as rigid blocks (Tapponnier *et al.* 1982; Peltzer & Tapponnier 1988; Avouac & Tapponnier 1993). Recently the relative importance of these two modes has been tested through velocity field determination of the Asian continent using the Holocene geological record and geodetic information from GPS data (Holt *et al.* 1991, 1995; Peltzer & Staucier 1996; England & Molnar 1997; Wang *et al.* 2001).

An alternative method for evaluating the style of deformation within the Asian continent is provided by palaeomagnetism. Post-Cretaceous clockwise rotation of crustal blocks is observed over a broad area that extends from eastern Tibet through South China to Indochina (Otofuji *et al.* 1990, 1998; Huang *et al.* 1992; Funahara *et al.* 1992, 1993; Huang & Opdyke 1993; Haihong *et al.* 1995; Sato *et al.* 1999, 2001; Yang *et al.* 2001). Large-scale southeastward displacement is suggested for the Indochina Block and the Simao Block (Yang & Besse 1993; Yang *et al.* 1995, 2001; Sato *et al.*

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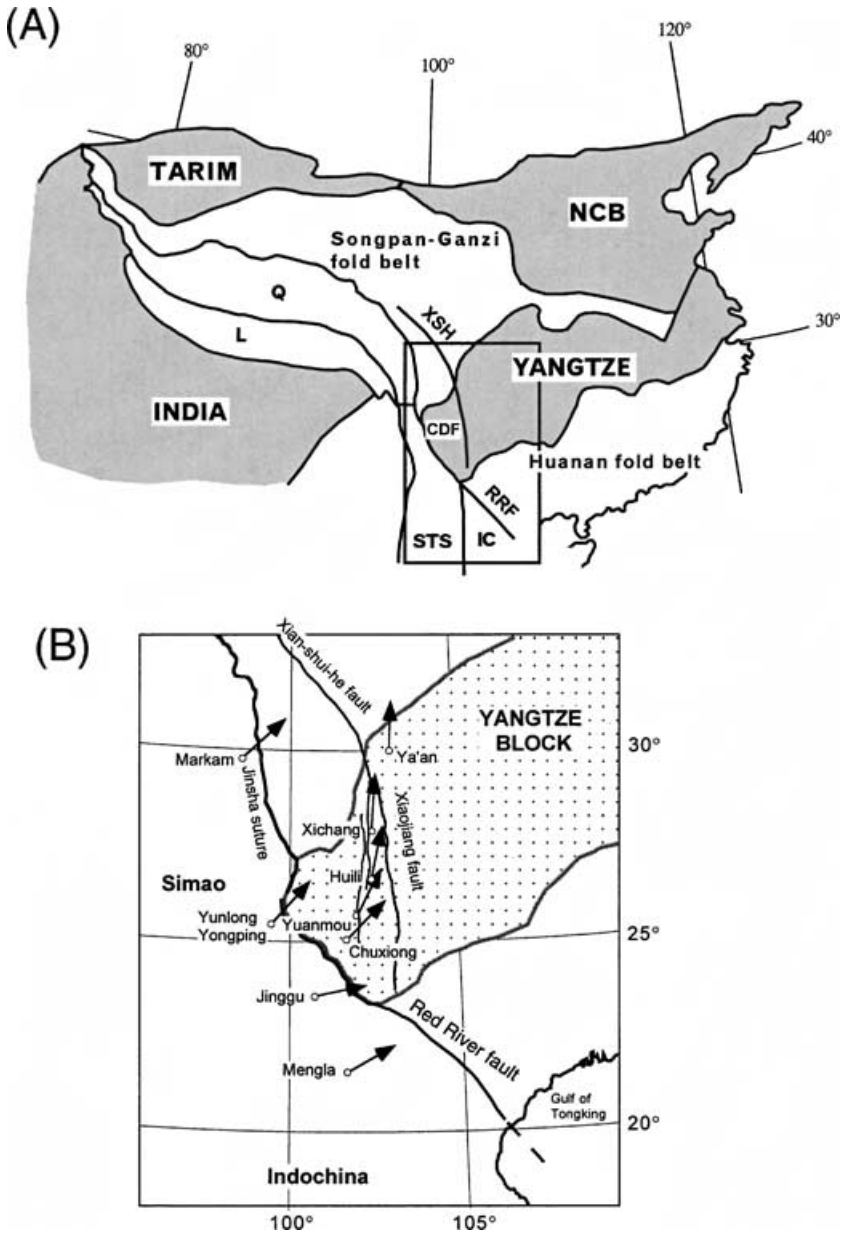


Figure 1. (a) Tectonic map of southeast Asia: NCB, North China Block; Q, Qiangtang Terrane; L, Lhasa Terrane; STS, Shan-Thai and Simao terranes; IC, Indochina Block; CDF, Chuan Dian fragment; XSH, Xianshuihe–Xiaojiang fault system; RRF, Red River Fault. (b) Location map of the study area. Main locations of Cretaceous palaeomagnetic data are available in the Sichuan Basin, and Shan-Thai and Simao terranes. Arrows indicate horizontal projections of palaeomagnetic remanent directions for the studied areas; Ya'an (Otofuji *et al.* 1990; Enkin *et al.* 1991), Xichang (this study), Huili (Huang & Opdyke 1992), Yuanmou (Otofuji *et al.* 1998), Chuxiong (Funahara *et al.* 1992), Markam (Otofuji *et al.* 1990; Huang & Opdyke 1993), Yunlong (Sato *et al.* 1999), Yongping (Funahara *et al.* 1993), Jinggu (Huang & Opdyke 1993; Haihong *et al.* 1995) and Mengla (Huang & Opdyke 1993).

1999). Palaeomagnetic data from the Yangtze, Indochina and Simao blocks offer an opportunity to better understand the intracontinental deformation of the Asian continent.

The present study focuses on the Chuan Dian fragment of the Yangtze Block, which is located at the southeastern margin of the Tibetan Plateau (Fig. 1). This fragment, bounded by the Xianshuihe–Xiaojiang and the Red River fault systems, is experiencing a southward movement at present (Chen *et al.* 2000). Analysis of earthquake data supports the southward displacement of this fragment (Holt *et al.* 1991; Chen *et al.* 1994). To detect southward displacement of this fragment, we carried out palaeomagnetic sampling of Cretaceous red sandstones near Xichang (27.9°N, 102.3°E). This study presents more reliable results than data reported earlier based on a very small number of samples (Zhu *et al.* 1988; Liu *et al.* 1999).

2 GEOLOGICAL SETTING AND SAMPLING

The Yangtze Block is one of the cratons that form the mainland of eastern Asia, and occupies much of the southern part of China (Fig. 1). Towards the southwest, this block is separated from the Shan-Thai, Simao and Indochina blocks by the Red River Fault. Although this fault with left-lateral motion was mainly active as

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one of the major strike-slip faults in eastern Asia between 35 and 17 Ma (Lacassin *et al.* 1997), right lateral slip started on it around 5 Ma (Allen *et al.* 1991).

The southwest part of the Yangtze Block is separated from the main part of the craton by the Xianshuihe–Xiaojiang fault system and forms a lens-shaped crustal fragment, the Chuan Dian fragment (Kang-Diang Block). Xichang (27.9°N, 102.3°E) is located on the northern part of the Chuan Dian fragment.

According to the Bureau of Geology and Mineral Resources of Sichuan Province (Bureau of Geology and Mineral Resources of Sichuan Province 1991), Cretaceous red beds are distributed in the Xichang area. This Cretaceous sequence includes the Lower Cretaceous Feitianshan Formation consisting of fluvio-lacustrine sandstones and mudstones with sandy conglomerates, which disconformably rests over the Upper Jurassic Guangou Formation. Because the lower part yields an ostracod assemblage *Minheella–Pinnocypridea*, and the upper part *Cypridea–Latonia*, Berriasian to Barremian ages are assigned to this formation. The Upper Cretaceous Xiaoba Formation disconformably overlies the Feitianshan Formation as well as the Jurassic strata, and disconformably underlies the Lower Tertiary Leidashu Formation. The Xiaoba Formation is made of lacustrine-facies sandstones and mudstones intercalated with carbonate and evaporite strata. The Coniacian to Maastrichtian is assigned on the basis of the abundant occurrence of ostracods, such as *Cypris* sp., *Eucypris* sp. and *Cyprinotus* sp., and charophytes such as *Obtusochara* sp. and *Sphaerochara* sp. Folding in this area took place during the Middle to Late Palaeogene, after the deposition of the Leidashu Formation, because almost horizontal Neogene strata (Xigeda Formation) overlie the deformed Palaeogene Leidashu Formation.

We collected red sandstones for palaeomagnetic purposes from both the Lower Cretaceous Feitianshan Formation and the Upper Cretaceous Xiaoba Formation in 1994. Sampling was carried out in two synclines, that is, 12 sites from the Dadeli Syncline (27°43.0′–43.9′N, 102°11.9′–14.9′E) and 21 sites from the Mishi Syncline (27°51.1′–52.9′N, 102°19.8′–33.3′E) (Fig. 2). The Dadeli Syncline is located at about 10 km south of Xichang city. The fold axis has a north–south trend and north plunge. Based on strikes and dips of bedding at opposite limbs, the intersection of the bedding planes from two limbs gives a possible fold axis that trends N19.6°E and plunges 23.3° toward the north. Dips of the bedding at the sampling sites range between 32° and 58°. The Mishi Syncline is broad across

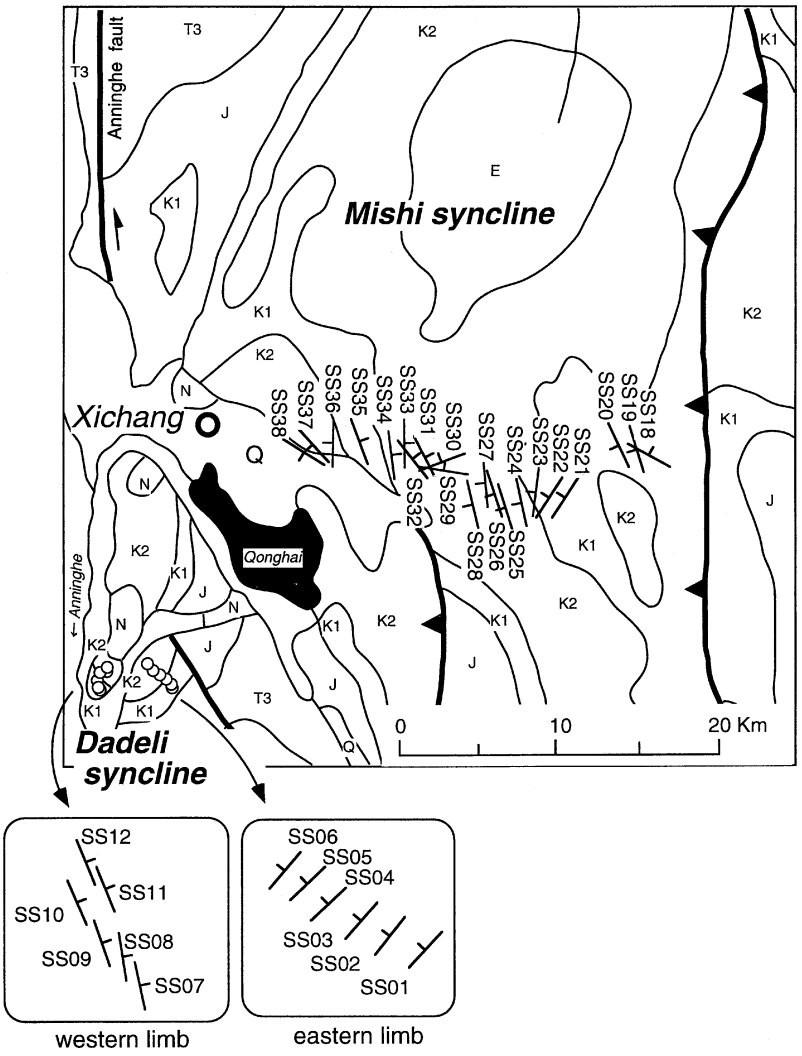


Figure 2. Geological map of the study area. Sampling sites are shown by open circles for the Dadeli Syncline and strike/dip orientations of strata for the Mishi Syncline. The bedding attitude at each sampling site from the Dadeli Syncline is shown in the inset. Black and barbed lines indicate faults and thrusts, respectively. T3, Late Triassic; J, Jurassic; K1 and K2; Early and Late Cretaceous; N; Neogene; Q, Quaternary.

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several tens of kilometres, and spreads widely east to Xichang city. This syncline contains small folds with a roughly north–south trend but does not plunge on average. The bedding attitude is fairly shallow and the angle of dip ranges between 3° and 40° (17° ± 8°).

Five to nine hand samples, oriented with a magnetic compass, were collected at each site covering an area up to 20 m. The bedding attitudes were measured with a magnetic compass at each site. The present geomagnetic declination value of 359.0° was evaluated using the Definitive Geomagnetic Reference Field of 1994 (IAGA Division V, Working Group 8 1995). Geographical coordinates of the sampling sites were determined by a portable navigational instrument (Sony Pyxis).

3 LABORATORY PROCEDURE AND ANALYSIS

Specimens (25 mm diameter, 22 mm high) were prepared for palaeomagnetic measurements in the laboratory. Natural remanent magnetization (NRM) was measured with an ScT cryogenic magnetometer. All the samples were subjected to stepwise thermal demagnetization up to 710°C. Results for each specimen were plotted on orthogonal vector diagrams (Zijderveld 1967) to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis (Kirschvink 1980) was used to estimate the component directions. Mean directions (Table 1) were calculated using Fisher statistics (Fisher 1953).

Progressive acquisitions of isothermal remanent magnetization (IRM) were performed up to a maximum field of 9 T using a 2G pulse magnetizer. Thermal demagnetization of the composite IRMs (2.65 T, 0.40 T and 0.12 T along each of three perpendicular axes) was carried out for detecting unblocking temperatures (Lowrie 1990). In addition, small fragments of specimens were subjected to thermomagnetic analysis for identifying ferromagnetic minerals in the rocks. The thermomagnetic analyses were carried out using an automatic recording thermomagnetic balance in air in a magnetic field of 0.85 T. Measurements of anisotropy of magnetic susceptibility (AMS) were made on specimens from the Dadeli and Mishi synclines using a Kappabridge KLY-3 apparatus. AMS data were analysed following the method of Tarling & Hrouda (1993). The principal susceptibilities (maximum $= K_1$, intermediate $= K_2$, minimum $= K_3$) are calculated for each specimen. The shape of magnetic fabric is represented by the shape parameter (T) and the degree of anisotropy (Pj).

4 ROCK MAGNETISM

A reversible cooling curve from thermomagnetic analysis demonstrates a very stable magnetic behaviour (Figs 3a and b). The Curie temperature of about 675°C indicates the predominance of stable haematite. IRM acquisition curves also show that saturation is not achieved up to the maximum field of 9 T (Fig. 3c), indicating the presence of high-coercivity magnetic minerals. Thermal demagnetization of the composite IRMs indicates an intensity drop at about 690°C, suggesting the presence of haematite (Figs 3d and e). Petrographic observation of the polished thin sections reveals that a detrital specular haematite is the predominant magnetic phase in the red sandstones (Fig. 4). These results thus suggest a predominance of haematite as the magnetic carrier in the studied samples.

AMS data for the Dadeli Syncline indicate that an original sedimentary fabric is slightly deformed (Fig. 5). In the *T–Pj* plots, the values of the degree of anisotropy concentrate around 1.02, while the values of the shape parameter reveal a large variation, indicating that the sedimentary oblate fabric has been modified. In geographical coordinates, maximum axes of most specimens form a moderate cluster around a northerly direction and inclined downwards by about 30°. Minimum and intermediate axes of susceptibility directions form a girdle around the clustered maximum axes. The direction of the cluster of maximum axes is consistent with the plunging of the fold axis expected from the geological structure (trend of N19.6°E and plunge of 23.3° toward the north).

This pattern does not have a sedimentary origin, but is interpreted as resulting from the development of folds in this tectonically deformed area. The maximum axis of the susceptibility direction is subparallel to the fold axes, while a semi-girdle distribution of minimum and intermediate axes appears from horizontal to very steep inclinations (Sagnotti *et al.* 1998; Pares *et al.* 1999). A sedimentary fabric in the Dadeli Syncline probably evolved to a composite of sedimentary and tectonic fabrics formed during folding. Alternatively, AMS data confirm the northward plunging of the fold axis for the Dadeli Syncline.

Considering the fold axis plunge (trend of N19.6°E and plunge of 23.3° toward the north) of the Dadeli Syncline, a two-step structural correction was applied to obtain the susceptibility direction in stratigraphic coordinates; the fold axis was unplunged to horizontal and subsequently each limb was unfolded. After the compound structural correction, the AMS results (Fig. 5) show a good example of the composite fabric with an originally sedimentary fabric overprinted by a compressional regime (Pares *et al.* 1999). The maximum axes are subhorizontal with cluster around a north–south direction, whereas the minimum and intermediate axes reveal a broad semi-girdle distribution. About 30 per cent of the minimum axes are grouped perpendicular to the bedding plane, indicating that the sedimentary fabric partly remains.

On the other hand, AMS data from the Mishi Syncline reveal a sedimentary fabric (Fig. 5). Oblate fabric is predominant and the degree of anisotropy extends up to 1.09. In stratigraphic coordinates, the minimum axes of susceptibility direction are grouped perpendicular to the bedding plane, whereas the maximum and intermediate axes form a girdle plane parallel to the bedding plane.

5 PALAEOMAGNETIC RESULTS

5.1 Dadeli Syncline

NRM intensities for the Feitianshan Formation (SS01–SS03, SS07–SS09) range from 4.5×10^{-3} to 1.9×10^{-2} A m^{-1}. NRM intensities for the Xiaoba Formation (SS04–SS06, SS10–SS12) are lower than those of the Feitianshan Formation, and range between 2.9×10^{-3} and 1.1×10^{-2} A m^{-1}. One to two NRM components were isolated by stepwise thermal demagnetization of each specimen (Figs 6a–c). The NRM component, which is removed between 200° and 500°C, is roughly identical to the present field direction in the study area. The high-temperature component was unblocked between 675°C and 710°C. The characteristic remanent magnetization (ChRM) of high unblocking temperature is identified in the samples from all 12 sites. The precision parameter of the site-mean directions ranges from 19.5 to 372.1.

Site-mean directions before tilt correction define two populations (Fig. 7). Six sites from the eastern limb of the syncline show northeast directions ($D = 46.7°$, $I = 53.3°$, $\alpha_{95} = 7.6°$), whereas the remaining six sites from the western limb reveal

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Table 1. Palaeomagnetic results for the Cretaceous redbeds of Xichang. Magnetization directions of the high-temperature component are listed. *n* and *N* are number of samples used for calculation and number of samples measured, respectively. Mean directions for the Feitianshan Formation (F), the Xiaoba Formation (X) and all sampling sites (both the Feitianshan and Xiaoba Formations) are calculated on the basis of palaeomagnetic directions presented as the value at the representative point of Xichang (27.9°N, 102.3°E). *D* and *I* are declination and inclination respectively; *k* is the estimate of the Fisherian precision parameter (Fisher 1953); α_{95} and A_{95} are the radii of cone of 95 per cent confidence. Strikes of bedding are values before subtraction of local geomagnetic declination. A value of 359.0° is calculated for the present local geomagnetic declination using the Definitive Geomagnetic Reference Field of 1994 (IAGA Division V, Working Group 8, 1995).

	Locality				*In situ*		Tilt corrected				Bedding	
Site name	Lat. (N)	Long. (E)	Formation	*n*/*N*	*D* (°)	*I* (°)	*D* (°)	*I* (°)	*k*	α_{95}(°)	Strike (°)	Dip (°)
Dadeli Syncline												
SS01	27°43.8′	102°14.9′	F	8/8	43.9	50.3	9.1	28.7	104.4	5.4	223	53
SS02	27°43.8′	102°14.9′	F	8/8	56.2	54.5	5.7	43.4	372.1	2.9	218	48
SS03	27°43.9′	102°14.8′	F	9/9	57.2	52.2	9.3	39.0	230.6	3.4	221	52
SS04	27°43.8′	102°14.9′	X	7/8	27.6	48.5	8.2	29.3	26.4	12.0	226	36
SS05	27°43.8′	102°14.8′	X	9/9	34.2	58.8	359.5	21.8	19.5	12.0	226	58
SS06	27°43.9′	102°14.8′	X	5/8	61.7	51.7	11.0	43.4	33.9	13.3	220	50
SS07	27°43.0′	102°11.9′	F	5/5	338.0	59.4	14.5	46.5	127.3	6.8	338	32
SS08	27°43.0′	102°11.9′	F	9/9	341.4	56.8	13.5	43.4	90.4	5.4	338	32
SS09	27°43.1′	102°11.9′	F	8/8	329.7	58.1	9.4	48.0	168.2	4.3	337	33
SS10	27°43.1′	102°12.0′	X	5/8	327.0	44.4	1.4	43.9	57.5	10.2	348	40
SS11	27°43.1′	102°12.0′	X	7/8	346.2	45.8	15.1	34.3	191.7	4.4	351	42
SS12	27°43.1′	102°12.0′	X	8/8	341.9	48.6	7.2	38.6	81.8	6.2	341	32
Mean	27°43′	102°13′		12	11.7	58.1			13.1	12.4		
				12			8.5	38.5	81.2	4.8		
Mishi Syncline												
SS18	27°52.4′	102°33.3′	F	6/8	357.8	56.8	351.5	45.9	52.9	9.3	241	12
SS19	27°52.6′	102°33.1′	F	7/8	358.4	46.1	355.3	46.9	801.0	2.1	163	3
SS20	27°52.8′	102°32.2′	F	8/8	25.9	29.6	10.6	48.1	55.9	7.4	155	26
SS21	27°51.2′	102°29.4′	F	8/8	359.5	49.3	346.7	38.5	105.2	5.4	213	17
SS22	27°51.2′	102°29.4′	F	8/8	357.1	59.8	340.0	47.9	161.8	4.4	213	17
SS23	27°51.1′	102°27.6′	F	8/8	14.5	44.4	354.8	43.9	134.3	4.8	187	20
SS24	27°51.1′	102°27.3′	X	8/8	22.0	46.2	355.7	57.7	86.7	6.0	161	23
SS25	27°51.3′	102°26.9′	X	8/8	16.2	41.8	352.3	51.2	240.7	3.6	163	24
SS26	27°51.5′	102°26.8′	X	7/8	19.9	37.9	0.4	49.4	125.2	5.4	162	23
SS27	27°51.7′	102°26.3′	X	7/8	19.5	30.6	3.6	38.0	53.1	8.3	175	24
SS28	27°51.2′	102°26.3′	X	6/8	15.2	31.0	359.8	39.6	108.4	6.4	167	23
SS29	27°52.5′	102°24.6′	X	7/8	15.2	53.0	13.9	43.1	63.4	7.6	279	10
SS30	27°52.5′	102°24.6′	X	8/8	10.0	37.9	7.8	32.7	35.6	9.4	249	6
SS31	27°52.5′	102°24.4′	X	8/8	4.7	45.7	12.8	39.1	83.8	6.1	333	11
SS32	27°52.5′	102°24.3′	X	7/8	358.0	37.7	2.4	32.6	31.6	10.9	322	8
SS33	27°52.8′	102°23.5′	X	7/8	6.7	44.8	21.8	40.0	32.1	10.8	359	17
SS34	27°52.6′	102°23.2′	X	8/8	15.5	34.0	21.1	29.6	56.4	7.4	353	10
SS35	27°53.6′	102°21.9′	F	8/8	350.0	45.5	12.9	34.0	104.8	5.4	340	29
SS36	27°53.2′	102°20.8′	X	8/8	20.5	40.1	349.1	43.1	90.5	5.9	181	34
SS37	27°52.9′	102°19.8′	X	8/8	354.6	37.1	347.7	42.7	20.0	12.7	138	10
SS38	27°52.9′	102°19.8′	X	8/8	348.6	37.0	343.2	43.4	354.0	3.0	122	9
Mean	27°52′	102°26′		21	8.7	42.7			47.3	4.7		
				21		0.7	42.9	51.1	4.5			
Mean	27.9°	102.3°										
Feitianshan				13	9.4	54.3			18.1	10.0		
				13			2.8	43.3	63.1	5.3		
Xiaoba				20	9.6	44.3			25.3	6.6		
				20			4.2	40.2	49.5	4.7		
All (Feitianshan+Xiaoba)				33	9.5	48.2			21.0	5.6		
				33			3.7	41.5	54.6	3.4		

Pole positions

	Lat. (°N)	Long. (°E)	A_{95}(°)
Feitianshan	86.6	237.0	5.8
Xiaoba	84.3	243.5	4.7
All	85.2	241.7	3.5

northwest directions ($D = 337.3°$, $I = 52.4°$, $\alpha_{95} = 6.8°$). Within each limb, the directions from the Feitianshan and the Xiaoba formations are statistically indistinguishable at the 95 per cent confidence level.

A two-step structural correction was applied to obtain palaeomagnetic directions in stratigraphic coordinates as in AMS; unfolding of each limb follows unplunging of the fold axis. After the compound correction, directions of the characteristic remanent

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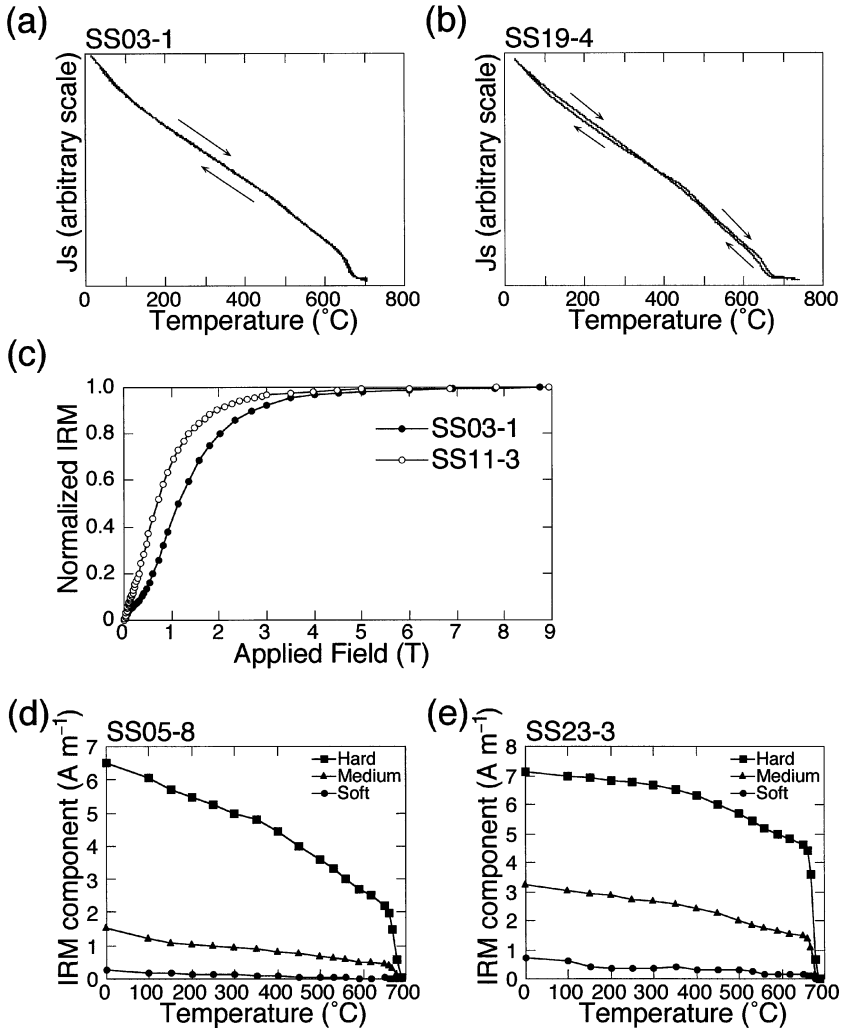


Figure 3. (a, b); High-field saturation magnetization (*Js*) versus temperature curves for samples from the Feitianshan Formation of the Dadeli Syncline (site SS03) and the Mishi Syncline (site SS19). Experiments were performed in air. The arrows indicate heating and cooling curves. (c) IRM acquisition curves for samples from the Feitianshan Formation (site SS03) and the Xiaoba Formation (site SS11) of the Dadeli Syncline. (d), (e) Thermal demagnetization of composite IRMs for samples from the Feitianshan Formation of the Dadeli Syncline (site SS05) and the Mishi Syncline (site SS23). IRMs of different dc fields (hard 2.65 T, medium 0.4 T, soft 0.12 T) were imparted to each of three perpendicular axes.

magnetization from either limb clustered in a single population (Fig. 7). When the McElhinny and McFadden fold tests (McElhinny 1964; McFadden 1990) were applied for a case of 100 per cent unfolding, positive tests were obtained at the 99 per cent confidence level. In the McFadden fold test (McFadden 1990), the calculated value (ξ_1) is 6.471 in *in situ* coordinates and 1.627 after tilt correction, while the critical value (ξ) is 5.624 at the 99 per cent confidence level. Applying the parametric bootstrap fold test of Tauxe & Watson (1994) to a data set after unplunging correction, the tightest grouping of data occurs between 74 per cent and 99 per cent unfolding in 95 per cent of the paradata set. Therefore, we conclude that the remanent magnetization was acquired critically before the folding. The mean direction after 100 per cent unfolding correction is $D = 8.5^\circ$, $I = 38.5^\circ$, $\alpha_{95} = 4.8^\circ$ ($N = 12$).

5.2 Mishi Syncline

NRM intensities range from 2.1×10^{-3} A m^{-1} to 3.5×10^{-2} A m^{-1}. During stepwise thermal demagnetization, samples from both the Feitianshan and Xiaoba formations show similar demagnetization behaviour. Like the Dadeli Syncline, one to two NRM components were observed. Samples having a univectorial component of magnetization were unblocked at about 680°C (Fig. 6f). For the samples with dual-component NRMs, the low-temperature component was removed between 200°C and 450°C, and the high-temperature component of magnetization was fully unblocked at about 680°C (Figs 6d, e). ChRM of the high-temperature component is identified in all 21 sites. Their directions are well grouped at each site; the precision parameter ranges from 20.0 to 801.0. The mean direction before fold correction for the Mishi Syncline is $D = 8.7^\circ$, $I = 42.7^\circ$, $\alpha_{95} = 4.7^\circ$ ($N = 21$) (Fig. 7).

The Mishi Syncline yields a very slight improvement in the concentration of the 21 site mean directions after tilt correction (Fig. 7). Although the precision parameter increases from 47.3 to 51.1, the fold test is inconclusive (McElhinny 1964; McFadden 1990). Five per cent stepwise unfolding for 11 sites (SS18–SS28) from the eastern subbasin shows that the precision parameter reaches a maximum at 100 per cent unfolding ($k_2/k_1 = 1.38$). The mean direction after 100 per cent unfolding correction for the Mishi Syncline is $D = 0.7^\circ$, $I = 42.9^\circ$, $\alpha_{95} = 4.5^\circ$($N = 21$).

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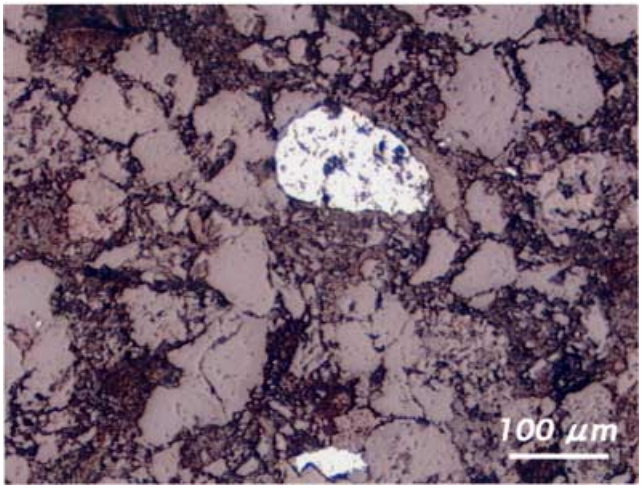


Figure 4. Photomicrograph under reflected light, showing the occurrence of a large specular haematite (bright). Sample SS03-2 at site SS03.

5.3 Xichang

We have successfully isolated a high-temperature component carried by haematite. In order to calculate a Cretaceous palaeomagnetic direction for Xichang, palaeomagnetic directions for the Dadeli Syncline after the compound correction are combined with those of the Mishi Syncline after 100 per cent fold correction. The overall mean direction, based on the 33 sites from both synclines, provides a characteristic direction of Cretaceous age for the Xichang area $D = 3.7^\circ$, $I = 41.5^\circ$, $\alpha_{95} = 3.4^\circ$ ($N = 33$) (Table 1), with a corresponding palaeopole at 85.2°N, 241.7°E ($A_{95} = 3.5^\circ$). This NRM direction passes the McElhinny fold test (McElhinny 1964) at the 99 per cent confidence level ($k_2/k_1 = 2.59$).

The occurrence of only normal polarity suggests that the characteristic remanent magnetization for the Lower Cretaceous Feitianshan Formation and the Upper Cretaceous Xiaoba Formation was acquired during the Cretaceous Long Normal Superchron (Gradstein *et al.* 1994; Opdyke & Channell 1996).

6 DISCUSSION

The Cretaceous palaeomagnetic pole presented here for Xichang (85.2° N, 241.7° E, $A_{95} = 3.5^\circ$) plots on the near side of the Cretaceous pole for the Sichuan Basin (76.8° N, 248.3° E, $dp/dm = 2.4^\circ/4.1^\circ$) with respect to the Xichang locality (Fig. 8a). The Cretaceous pole for the Sichuan Basin is based on previous palaeomagnetic results reported from three areas around Ya'an (30°N, 103°E)—Xinjin, Feizianguan and Guanyin (Otofuji *et al.* 1990; Enkin *et al.* 1991). These palaeomagnetic results (listed in Table 2) were obtained from continental redbeds which passed the fold test. Comparison of these two palaeomagnetic pole positions suggests that the Xichang area underwent southward displacement with respect to the Sichuan Basin.

The appearance of the available palaeomagnetic data suggests that the process of southward displacement is not only restricted to

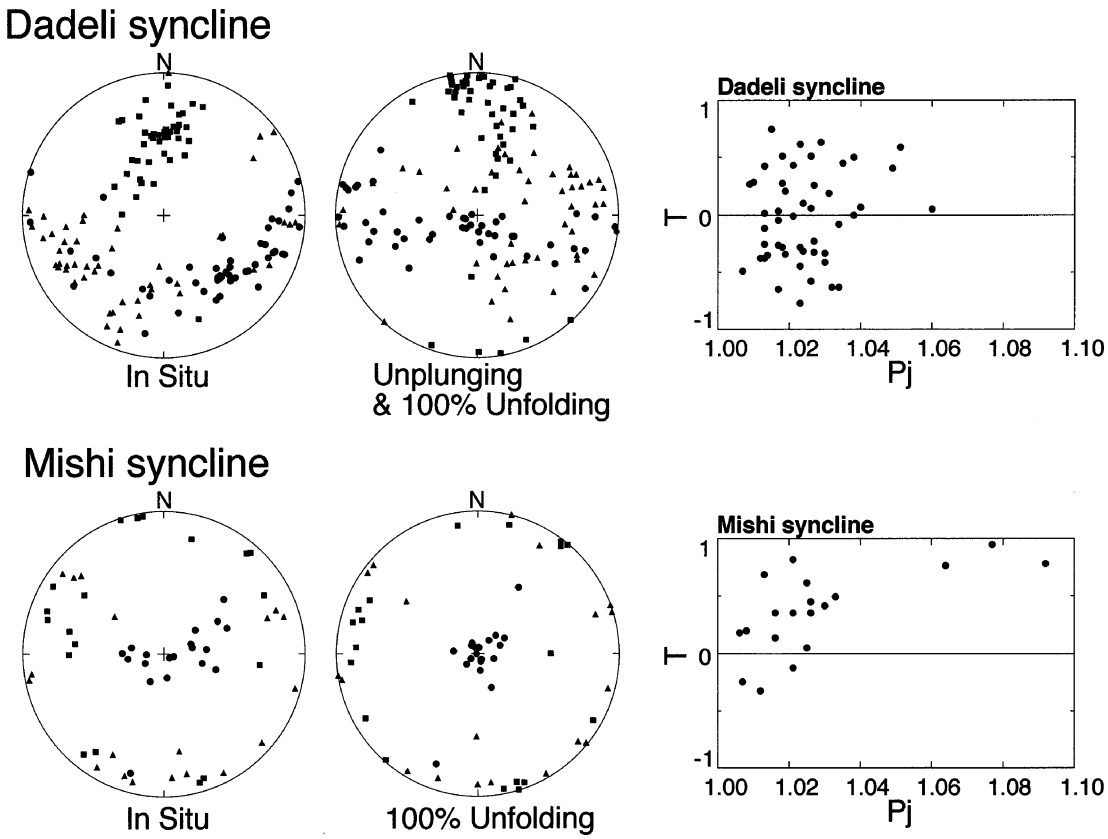


Figure 5. The principal directions of susceptibility for the Dadeli and the Mishi synclines together with plots of shape parameter (T) and the degree of anisotropy (Pj). The principal directions are projected onto the lower hemisphere (equal-area projection). Squares are the maximum susceptibility direction (K_1); triangles are intermediate susceptibility directions (K_2); circles are the minimum susceptibility directions (K_3).

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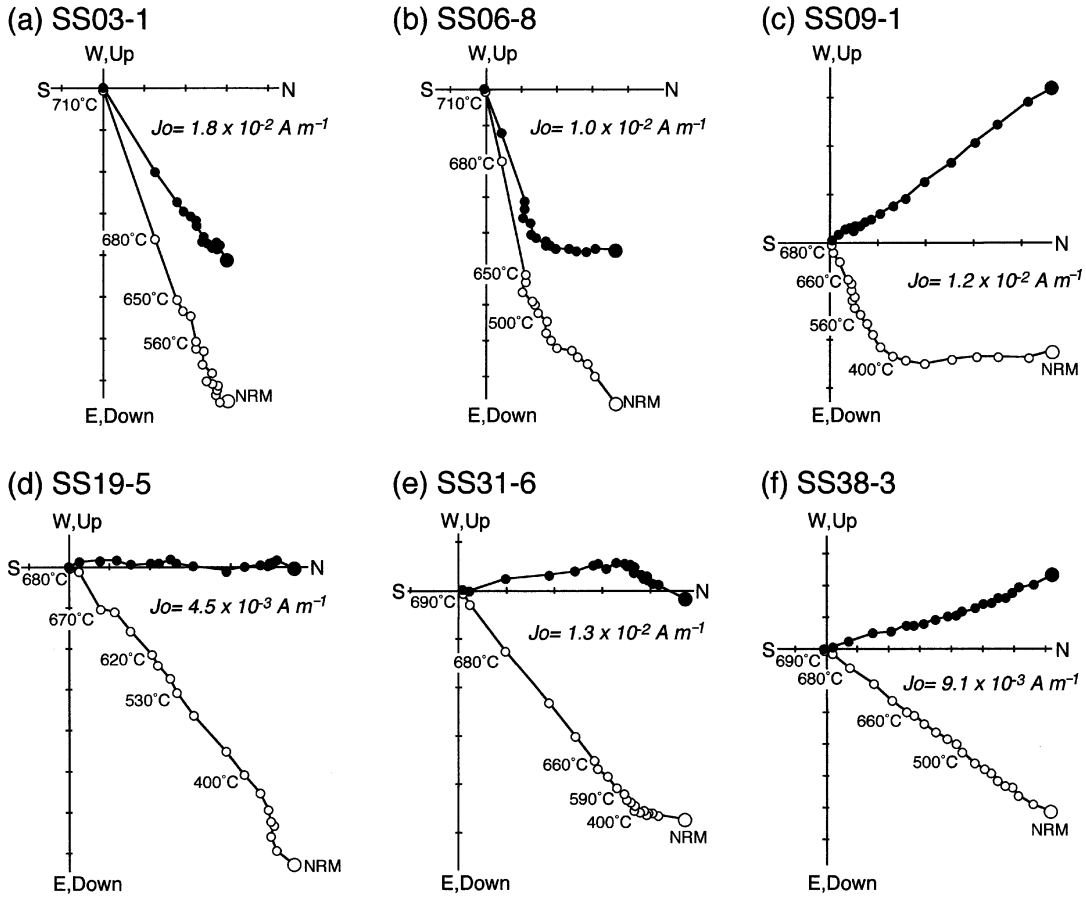


Figure 6. Orthogonal projections of magnetization in geographical coordinates in thermal demagnetization experiments. (a–c) Dadeli Syncline, (d–f) Mishi Syncline. Examples of the Feitianshan Formation are shown in (a), (c) and (d), and examples of the Xiaoba Formation are in (b), (e) and (f). Open and solid symbols show projection onto the vertical and the horizontal planes respectively.

Table 2. Cretaceous palaeomagnetic directions and poles from the Chuan Dian fragment and Sichuan Basin. N is number of sites used for calculation. α_{95} and A_{95} are the radii of cone of 95 per cent confidence. dp and dm are radii of the semi-axis of the ellipse of 95 per cent confidence along the site-to-pole great circle and perpendicular to the Great Circle, respectively. J3, the Late Jurassic; K1 and K2, the Early and Late Cretaceous, respectively.

[Chuan Dian fragment]					Palaeomagnetic direction			Pole position			
Locality	Age	N	Lat.(°)	Long.(°)	D (°)	I (°)	α_{95}(°)	Lat. (°N)	Long. (°E)	A_{95}(°)	References
Xichang	K1–K2	33	27.9	102.3	3.7	41.5	3.4	85.2	241.7	3.5	This study
Huili	K1	7	26.8	102.5	22.1	37.1	3.8	69.0	204.0	4.3	Huang & Opdyke (1992)
Huili	K2	18	26.5	102.4	8.1	38.8	6.6	81.9	220.9	7.1	Huang & Opdyke (1992)
Yuanmou	K2	21	25.9	101.7	26.9	35.9	3.6	64.6	199.6	3.3	Otofuji et al. (1998)
Chuxiong	K2	21	25.0	101.5	44.6	41.3	10.7	49.2	182.0	11.4	Funahara et al. (1992)
[Sichuan Basin]					Palaeomagnetic direction			Pole position			
Locality	Age	N	Lat.(°)	Long.(°)	D (°)	I (°)	α_{95}(°)	Lat. (°N)	Long. (°E)	dp(°)/dm(°)	References
Xinjin	K2	4	30.4	103.8	19.2	39.2	12.1	71.0	214.8	8.6/14.5	Enkin *et al.* (1991)
Ya'an	K1	9	30.1	103.0	2.1	30.7	11.3	76.3	274.5	11.1	Otofuji *et al.* (1990)
Feixianguan	J3–K2	26	30.0	102.9	2.0	34.2	3.6	78.6	273.4	2.4/4.1	Enkin *et al.* (1991)
Guanyin	J3–K2	14	29.1	104.6	15.8	32.7	7.8	71.7	229.1	5.0/8.8	Enkin *et al.* (1991)
Average of site grouping for Sichuan Basin											
Xinjin, Feixianguan, and Guanyin	J3-K2	44	30.0	103.0	7.9	34.4	3.6	76.8	248.3	2.4/4.1	Enkin *et al.* (1991)

the Xichang area but affects the whole Chuan Dian fragment. As shown in Fig. 8(a) and listed in Table 2, five Cretaceous palaeomagnetic poles are now available from the Xichang, Huili, Yuanmou and Chuxiong areas of this fragment (Funahara *et al.* 1992; Huang & Opdyke 1992; Otofuji *et al.* 1998). All these poles are arranged along a small circle through the Xichang pole centred on the Xichang locality. The small circle occurs on the near side from the pole of the Sichuan Basin with respect to the Xichang locality. The distribution of poles from the Chuan Dian fragment implies a variable amount of vertical-axis rotation for each locality and southward displacement of all five areas with respect to the Sichuan Basin.

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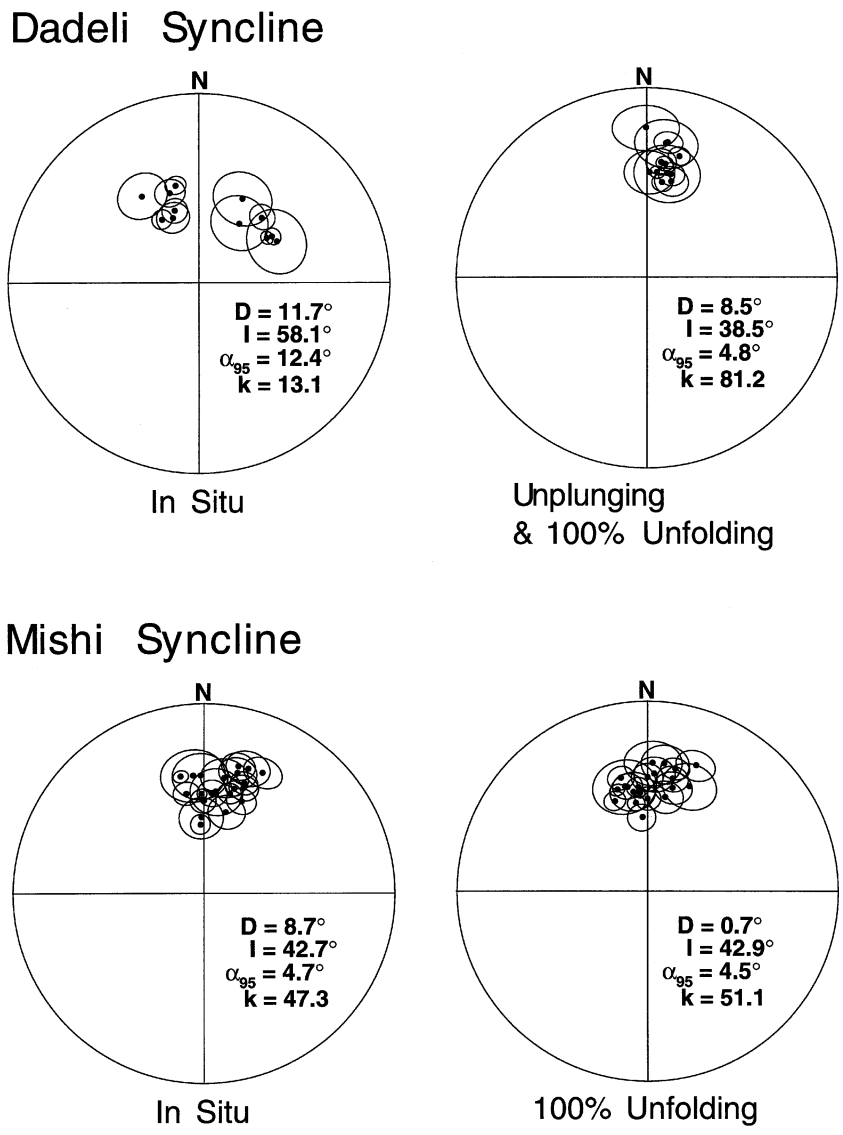


Figure 7. Equal-area projections of site mean directions with 95 per cent circles of confidence for the Dadeli and Mishi synclines. Solid symbols are lower hemisphere projections.

To estimate the magnitude of southward displacement, the observed palaeolatitudes of the Chuan Dian fragment are compared with those expected from the Cretaceous palaeomagnetic pole of the Sichuan Basin (Fig. 8b). All five observed palaeolatitudes from the Chuan Dian fragment are higher (5.0° to 9.9°) than the expected values. The arithmetic mean of the variance for five areas gives southward displacement of 6.7°, with standard deviation of 2.0°. The uncertainty (Δ) in the palaeolatitude difference at 95 per cent confidence level is calculated as $\Delta = t_c\, S/N^{-1/2}$, where N (= 5) is the number of areas, S is the standard deviation of the arithmetic mean and t_c is the confidence coefficient at the 97.5 per cent point on the Student's distribution with $N - 1$ degrees of freedom. Using the 95 per cent confidence of the expected palaeolatitude ($dp = 2.4°$), the uncertainty increases up to 3.5°. Our final estimate is $6.7° \pm 3.5°$ change in latitude. We therefore conclude that the Chuan Dian fragment has undergone southward displacement of more than 350 km with respect to the Sichuan Basin since the Cretaceous.

Neogene southward displacement of this fragment is also supported by geological and geodetic observations. GPS measurements clearly show southward displacement of this fragment with respect to the South China Block at the present rate of 5.3 ± 3.1 mm yr^{-1} (Chen *et al.* 2000). The Xianshuihe–Xiaojiang fault system separates the Chuan Dian fragment from the Sichuan Basin (Figs 1 and 8b). Movement observed along this left-lateral fault system suggests a southeastward displacement of about 60 km for the Chuan Dian fragment during the last 4 Myr. Southward movement of 15 ± 5 mm yr^{-1} to 5 mm yr^{-1} has been reported by Allen *et al.* (1991) and Wang *et al.* (1998) for this fragment. Roger *et al.* (1995) reported that left-lateral motion along this fault system has already begun by 12 Ma.

India first collided with Asia at about 50 Ma but indentation of the former into the latter is progressing even today (Rowley 1996; Wang *et al.* 2001). A similar tectonic situation has been maintained from the Palaeogene to the present in the eastern Tibetan region. Assuming that the Chuan Dian fragment commenced southward motion after the initial India–Asia collision at 50 Ma, its displacement rate is estimated at 15 ± 8 mm yr^{-1}. This is consistent with the rate of the Miocene displacement expected from the geological viewpoints. Although no clear geological evidence is reported, present palaeomagnetic study suggests that southward movement of the Chuan Dian fragment has been continuing since the time of collision.

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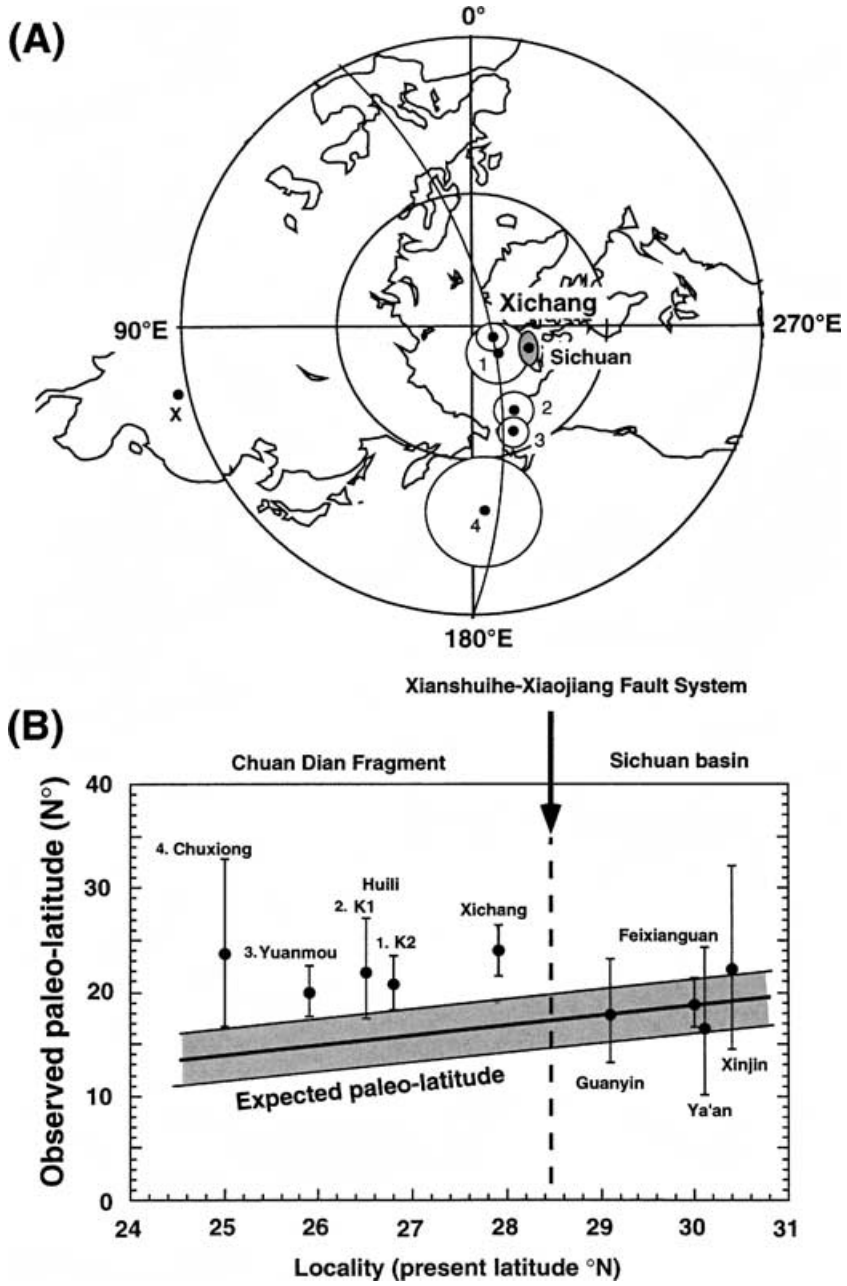


Figure 8. (a) Polar stereographic projection of palaeomagnetic poles (with associated circles or ellipse of 95 per cent confidence level) for Cretaceous redbeds from the Chuan Dian fragment and the Sichuan Basin (shaded). Small circle centred on Xichang (x) (27.9°N, 102.3°E) is projected through the Xichang pole. 1, K2 of Huili; 2, K1 of Huili; 3, Yuanmou; 4, Chuxiong (Funahara *et al.* 1992; Huang & Opdyke 1992; Otofuji *et al.* 1998). (b) Palaeolatitude versus age for five palaeomagnetic localities in the Chuan Dian fragment and for four palaeomagnetic localities in the Sichuan Basin. These localities are arranged on the present latitude. Error bar shows uncertainty at 95 per cent confidence level. Expected palaeolatitude and its 95 per cent confidence interval are shown by the black line and shaded area, respectively. The expected palaeolatitudes are values for the present latitude along the present longitude of 120°E calculated from the Cretaceous palaeomagnetic pole of the Sichuan Basin (76.8° N, 248.3° E, $dp = 2.4°$, $dm = 4.1°$) (Enkin *et al.* 1991).

The eastern Tibetan region constitutes a widely distributed sinistral shear zone (Ma 1986; Avouac & Tapponnier 1993; Lacassin *et al.* 1997). Due to the regional tectonic regime, the Chuan Dian fragment has been displaced along the Xianshuihe–Xiaojiang fault system, while the Indochina Block experienced southward displacement along the sinistral Red River Fault after the Cretaceous (Leloup *et al.* 1995; Sato *et al.* 1999, 2001; Yang *et al.* 1995, 2001). Laboratory experiments reveal that convergence between India and Asia is partly absorbed by the lateral extrusion of continental size blocks along major strike-slip faults (Tapponnier *et al.* 1982; Peltzer & Tapponnier 1988; Davy & Cobbold 1988). The post-Palaeogene southward movement of the Chuan Dian fragment is thus ascribed to the process of extrusion which effected several crustal blocks within the Asian continent due to indentation of India.

The Xianshuihe–Xiaojiang fault system probably played an important role in controlling southward displacement of the Chuan Dian fragment. Advection of lower crustal flow is another important factor in southward movement of the Chuan Dian fragment.

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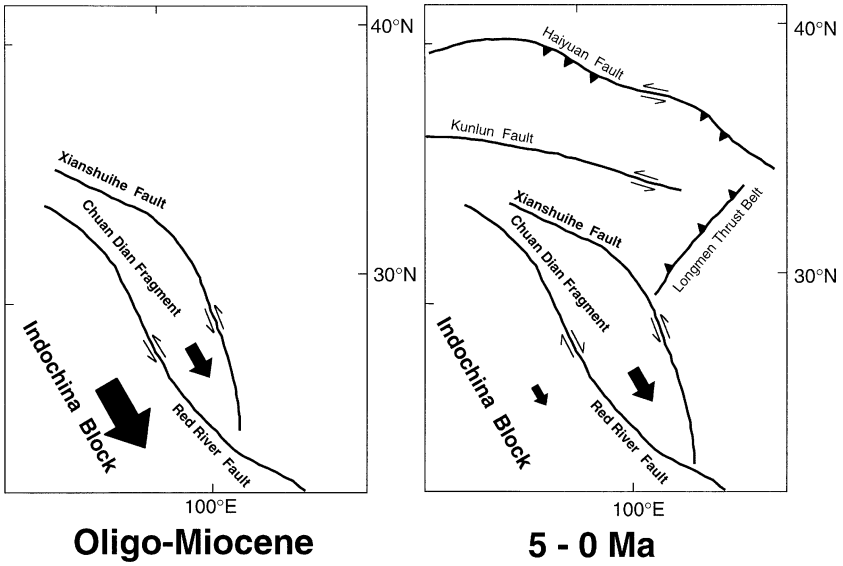


Figure 9. Southward displacement aspect of the Chuan Diang fragment. In Oligo-Miocene times, movement of Indochina is larger than that of the Chuan Dian fragment. During this period, the Indochina Block was separated from the Chuan Dian fragment by the left-lateral Red River Fault. However, by about 5 Ma, southward displacement of the Chuan Dian fragment overtook the movement of the Indochina Block. Due to this significant change, the left-lateral slip along Red River Fault changed into right-lateral movement.

As suggested by Clark & Royden (2000), lower crustal material squeezed southeastwards into the Indochina Block from the Tibetan Plateau. The Sichuan Basin, which is the oldest basement of the Yangtze Craton (Li 1998), obstructed this flow due to its deep and strong root, and the flow eventually made a path along the western margin of the basin. The Xianshuihe–Xiaojiang fault system may represent a border between the strong crustal embankment of the Sichuan Basin and the stream of flow.

Taking into consideration the concurrent southward displacement of the Chuan Dian fragment and Indochina Block, the relative rate of southward displacement between these two terranes can be predicted in the sense of lateral motion along the Red River Fault (Fig. 9). Left-lateral slip along the Red River Fault occurred in Oligo-Miocene times, whereas a change to right-lateral movement has been registered at about 5 Ma (Leloup *et al.* 1995; Lacassin *et al.* 1997). This implies that Indochina moved southward faster than the Chuan Dian fragment until about 5 Ma. At about 5 Ma, the rate of displacement for the Indochina Block slowed down, while southward displacement of the Chuan Dian fragment was still in progress. Larger GPS velocity vectors are observed in the Chuan Dian fragment than in the Indochina Block (Wang *et al.* 2001).

Clockwise palaeomagnetic rotation is only restricted to the southern part of the Chuan Dian fragment (Fig. 1b and Table 2), because a northerly declination from the present study indicates a negligible amount of palaeomagnetic rotation in the Xichang area ($D = 3.7^\circ$). The declination value increases from 4° to 45° as the latitude decreases from 27.9°N to 25.0°N. A systematic increase in declination toward the southern part of the Chuan Dian fragment is clearly manifested by palaeomagnetic pole data (Fig. 8a). Since recent palaeomagnetic studies suggest that crustal rotation due to fault motion is limited to within a few kilometres of the fault zone (Otofuji *et al.* 1995; Piper *et al.* 1997), the Cenozoic left-lateral strike-slip motion of the Xianshuihe–Xiaojiang fault system and Red River Fault probably played only a small role in crustal rotation of this area. We interpret tectonic deformation in the southern part of the Chuan Dian fragment to be a local phenomenon. This type of behaviour in crustal rotation indicates that tectonic stresses responsible for deformation affected only the southern part of the studied fragment whereas the northern part has not been influenced. It seems a reasonable assumption that this fragment jostled against the Yangtze and Indochina blocks during its southward displacement. As a result, only an apical part of this fragment, located between the Xianshuihe–Xiaojiang and the Red River fault systems, was probably deformed during its extrusion from the north.

7 CONCLUSION

Palaeomagnetic study of Early to Late Cretaceous red sandstones from the Chuan Dian fragment shows that this fragment experienced a southward displacement of $6.7^\circ \pm 3.5^\circ$ with respect to the Sichuan Basin after the Cretaceous. The set of declination data indicates that only the southern part of this fragment was subjected to clockwise tectonic rotation. Extrusion dynamics in the Asian continent, due to its collision with India, brought about the southward displacement of the Chuan Dian fragment, which later bumped against the Yangtze and Indochina blocks. The Chuan Dian fragment, separated by narrow zones of the Xianshuihe–Xiaojiang and the Red River fault systems, generally behaved as a coherent block during its southward displacement, but its apical part exceptionally was subjected to rotational deformation in a clockwise sense.

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